



Question Set

Question 1

Let X and Y be risk-factor shocks with joint density $f(x,y)$. Derive the marginal density of X and the conditional density of Y given $X=x$.

Question 2

For jointly normal

$$X, Y$$

with means 0, volatilities

$$\sigma_X, \sigma_Y$$

and correlation

$$\rho$$

state

$$\mathbb{E}[Y | X = x]$$

and

$$\text{Var}(Y | X = x)$$

Question 3

A portfolio loss is

$$L = 3X + 5Y$$

where X and Y have volatilities 2 and 4 and correlation 0.6. Compute

$$\text{Var}(L)$$

Question 4

Explain tail dependence and why ordinary correlation can miss it.

Question 5

Two default indicators have $P(D1=1)=0.04$, $P(D2=1)=0.06$, and $P(D1=D2=1)=0.01$. Test independence and interpret.

Question 6

Derive the covariance matrix of a two-factor linear PnL vector

$$\Delta\Pi = BX$$

where

$$X$$

has covariance

$$\Sigma_X$$

Question 7

If X, Y are independent continuous variables, prove $f_{\{X,Y\}}(x,y) = f_X(x)f_Y(y)$. What is the risk interpretation?

Question 8

A Gaussian copula uses normal latent variables to join non-normal marginals. Explain the construction in three steps.

Question 9

For bivariate normal shocks, why is a linear portfolio PnL exactly normal but an option PnL not exactly normal?

Question 10

Define covariance matrix positive semidefiniteness and link it to joint distributions.

Question 11

Explain why diversification benefit is nonlinear in correlation near high correlations.

Question 12

A joint model fits marginal VaR for each desk but underestimates portfolio VaR. What model weakness does that suggest?